

Exam Date & Time: 15-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Advanced Software Engineering [IT_4417]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the Cleanroom Software Engineering process. How does it differ from conventional testing strategies in ensuring defect prevention and reliability? (5)
 - A)
 - B) Evaluate the strengths and limitations of formal specification languages in modeling concurrent systems. Discuss their impact on scalability and correctness. (5)
- 2) Analyze a hypothetical failure scenario in an avionics autopilot system. Identify safety-critical components, potential failure modes, and propose how a safety case structure (claims, arguments, and evidence) can be developed. (5)
 - A)
 - B) Critique the effectiveness of safety cases in certifying safety-critical systems such as automotive braking or flight control. Support your evaluation with suitable industry examples. (5)
- 3) Compare the lifecycle activities of safety-critical systems with those of conventional software systems. What unique verification, validation, and certification challenges arise in safety-critical development? (5)
 - A)
 - B) Evaluate the suitability of component composition techniques in dynamic runtime environments. Discuss how interface contracts and dependency management affect adaptability and performance. (5)
- 4) Evaluate the effectiveness of Software Product Lines in reducing development cost and time-to-market. Discuss factors affecting their industrial adoption. (5)
 - A)
 - B) Explain how Software Product Lines enable systematic reuse across multiple applications. Analyze how core asset management supports flexibility and scalability. (5)
- 5) Evaluate the contribution of timing analysis in ensuring predictability and schedulability of real-time applications. Illustrate with an example from an embedded control system (5)
 - A)

- B) Assess the role of Artificial Intelligence (AI) in automating classical software engineering tasks such as code generation, defect prediction, and fault localization. (5)
Discuss its implications for future software development practices.

-----End-----

Event Name		Subject	
058-END SEMSTER-ODD-2025		IT_4417:Advanced Software Engineering	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	20
4	4	CO 4	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		IT_4417:Advanced Software Engineering	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	4	20(40%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)